Synoptic λ-Coherence

A Modal Bible Approach to the Synoptic Problem

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Abstract

The Synoptic Problem asks how the Gospels of Matthew, Mark, and Luke are literarily related. Classical solutions (Two-Source, Farrer, Griesbach, Augustinian, etc.) compare surface text: wording, order, and shared or unique episodes. This paper proposes a different layer of analysis: **λ-coherence**.

Using *iota Verbum*'s "Modal Bible" framework, each Synoptic pericope is encoded as a machine-readable λ-document whose core is not the exact wording, but a **modal invariant** of what is being asserted about Jesus, his identity, and the effects of his actions. We then compare these invariants across Matthew, Mark, and Luke using a simple similarity measure, and model the main Synoptic hypotheses in a shared modal grammar.

The result is not a final "solution" to the Synoptic Problem, but an **operational test harness**: a way to state hypotheses as logical constraints, encode pericopes as executable invariants, and see which patterns of dependence (Markan priority, Q-like layers, or Farrer-style reuse) yield the most stable λ-coherence with the least drift. This same modal engine later powers iota Verbum's trust infrastructure for credit, healthcare, and law. The New Testament becomes its first and most demanding test corpus.

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1. Introduction

The Gospels of **Matthew, Mark, and Luke** share so much material that they can be laid out in parallel columns and "seen together" (*syn-optic*). At the same time, they differ in wording, order, emphasis, and unique content. This tension has generated the classical **Synoptic Problem**:

How are Matthew, Mark, and Luke literarily related, and what sources or redactional patterns best explain their agreements and differences?

Traditional approaches compare **surface text** (Greek words, clauses, and narrative sequence). By contrast, the *Modal Bible* project treats each pericope as a structured **λ-document** encoding:

- **Identity invariants** – what must be true of Jesus and the Kingdom in this passage.
- **Enacted episodes in time** – particular events, teachings, or encounters.
- **Observed effects** – what happens to people, institutions, and the story world.

All of this is expressed in a small **modal grammar** (□ / ◇ / Δ / ⊗Λ) that can be executed and analysed by machine.

This paper uses that grammar to define **Synoptic λ-coherence**:

Three pericopes (Matt–Mark–Luke) are λ-coherent if they share a stable invariant about Jesus' identity and mission, even when narrated differently.

We then:

1. Outline the classical Synoptic Problem in brief.
2. Describe the λ-encoding used for Matthew, Mark, and Luke.
3. Define a lightweight similarity metric over λ-invariants.
4. Present the first set of results from a sample of major Synoptic pericopes.
5. Show how key hypotheses (Markan priority, Q-like layers, Farrer) can be expressed and stress-tested in the same modal grammar.

The aim is modest but concrete: **make the Synoptic Problem executable**. Instead of only arguing in prose, we can turn hypotheses into code and inspect whether the resulting λ-corpus hangs together.

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2. The Classical Synoptic Problem (Very Brief)

For over a century, three families of solutions have dominated discussion:

- **Markan priority + Q (Two-Source Hypothesis)**
  Mark is written first; Matthew and Luke independently use Mark plus a second, hypothetical source "Q" for double tradition material (e.g. Beatitudes, Lord's Prayer), with their own special sources M and L for unique material.
- **Markan priority without Q (Farrer-type models)**
  Mark is first; Matthew uses Mark; Luke uses both Mark and Matthew. No Q is needed; Luke's overlaps with Matthew are direct dependence rather than a lost source.
- **Matthean-priority families (Augustinian / Griesbach)**
  Matthew is first; Mark abbreviates Matthew; Luke uses one or both. In some variants, Mark conflates Matthew and Luke.

Each model tries to explain three data types:

1. **Triple tradition** – passages shared by all three Gospels.
2. **Double tradition** – material shared by Matthew and Luke but not Mark.
3. **Single tradition** – material unique to one Gospel.

Classic arguments rely on vocabulary, style, order of events, and inferred editorial tendencies. The **Modal Bible** does not replace this work; instead, it adds a **structural layer**:

- not: "Who copied which wording from whom?"
- but: "Which Gospel's λ-invariant best functions as the **minimal common skeleton** that others expand or adapt?"

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3. The Modal Bible and λ-Documents

3.1 Core Modal Grammar

iota Verbum's theological core is captured in a simple chain:

[
□L ;\to; ◇E,L(t) ;\to; Δ(t) ;\to; ⊗Λ
]

Where:

- **□L** – Necessary Love / Law / Logos: who God is and what must hold.
- **◇E L(t)** – Enacted in time: the same love expressed in particular episodes *t*.
- **Δ(t)** – Effects: the real, accountable consequences in the world.
- **⊗Λ** – Lament tensor: how remaining gaps, harm, or "not-yet" are held faithfully.

In the Modal Bible, each pericope is a **λ-unit** that instantiates this pattern:

- A set of **invariants** (□…): what the text treats as non-negotiable.
- A series of **enactments** (◇…): actions or sayings of Jesus and others.
- A record of **effects** (Δ…): changed hearts, healed bodies, hardened opposition.
- A pointer to **laments / tensions** where the story is unresolved (⊗Λ).

This same grammar later generalises to institutional decisions (credit, healthcare, law), but its first and hardest domain is Scripture.

3.2 Data Model for a λ-Document

Concretely, a λ-document for a pericope (say, the Calming of the Storm) is stored as JSON:


```
{
  "gospel": "Matthew",
  "pericope_id": "Mat08_23_27",
  "pericope_label": "CalmingTheStorm",
  "reference": "Matt 8:23–27",
  "units": [
    {
      "unit_id": "Mat08_23_27.u1",
      "role": "identity",
      "statements": [
        {
          "id": "Mat08_23_27.s1",
```

"lambda_iv": "□ LordOfCreation(Jesus) → ◇ CommandsWindAndSea(Jesus) → Δ GreatCalm ∧ FearAndAwe(Disciples)"

}

]

}

]

}

The details vary, but the **key object** is lambda_iv: a single modal formula capturing the canonical invariant expressed in that pericope.

Equivalent λ-documents exist for Mark and Luke:

- Mar04_35_41.lambda.json for Mark 4:35–41
- Luk08_22_25.lambda.json for Luke 8:22–25

3.3 The Synoptic Alignment File

To compare across Gospels, we maintain a small alignment index:

{

"CalmingTheStorm": {

"matthew": "Mat08_23_27",

"mark": "Mar04_35_41",

"luke": "Luk08_22_25"

},

"BaptismOfJesus": {

"matthew": "Mat03_13_17",

"mark": "Mar01_01_13",

"luke": "Luk03_21_22"

},

...

}

Each **synoptic key** points to the pericope IDs in the three Gospels. A simple script then:

1. Loads each λ-document.
2. Extracts the lambda_iv formula for each.
3. Measures similarity between them.

That is the **λ-coherence** experiment.

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4. Method: Defining and Measuring λ-Coherence

4.1 Corpus for the First Experiment

For this initial paper, we focus on a **small but theologically central set** of Synoptic pericopes, including:

- Baptism of Jesus
- Temptation of Jesus
- Call of the first disciples
- Parable of the Sower
- Calming of the Storm
- Gerasene demoniac
- Jairus & bleeding woman
- Feeding of the 5,000
- Peter's confession
- Transfiguration
- Rich young ruler
- Servant-ransom saying
- Triumphal entry
- Last Supper & Gethsemane
- Crucifixion core
- Empty tomb

For each, we have:

- One λ-document in Matthew (MatXX_YY_ZZ)
- One in Mark (MarXX_YY_ZZ)
- One in Luke (LukXX_YY_ZZ), except where Luke omits the pericope.

4.2 Similarity Metric

To keep the first pass **transparent and reproducible**, we use a simple "edit distance-style" metric over the lambda_iv strings:

- Tokenise the lambda_iv formulas by splitting on whitespace.
- Use Python's difflib.SequenceMatcher to compute a ratio between 0 and 1, where:
    - **1.0** = identical sequences
    - **0.0** = no recognisable overlap

For each Synoptic key, we compute:

- sim(Matt, Mark)
- sim(Matt, Luke)
- sim(Mark, Luke)

This is intentionally modest. It does **not** yet use embeddings or deep semantics; it simply tests: *"When we encode pericopes by hand, which pairs of Gospels end up with more closely matching modal invariants?"*

4.3 λ-Coherence Definitions

We can now define:

- **λ-equivalence**:
  Two pericopes P and Q are λ-equivalent if their lambda_iv are identical, modulo variable renaming and trivial formatting.
- **λ-similarity**:
  For a threshold τ (e.g. 0.7), P and Q are λ-similar if sim(P, Q) ≥ τ.
- **λ-coherent triple**:
  A triple (Matthew, Mark, Luke) is λ-coherent if at least two pairs are λ-similar at τ, and all three entail a common invariant core when expressed in canonical normal form.

In practice, this first experiment simply reports the similarity scores and patterns. Later work can formalise normal forms and entailment.

4.4 Encoding Hypotheses in Modal Form

The standard hypotheses can be sketched in the same grammar.

Let:

- $\mathbf{M^a}$, $\mathbf{M^k}$, $\mathbf{M^l}$ = λ-corpora for Matthew, Mark, and Luke respectively.
- **Inv(X)** = set of invariants encoded for corpus X.
- **⊆** = logical subsumption: "every invariant in X is entailed by Y."

Then, for example:

- **Markan priority (λ-skeleton view)**
  Mark provides the **minimal λ-skeleton**; Matthew and Luke expand it.

[
H_{Mk} : Inv(M^{k}) ⊆ Inv(M^{a}) ;\wedge; Inv(M^{k}) ⊆ Inv(M^{\ell})
]

- **Farrer-style dependence (no Q)**
  Luke depends on both Mark and Matthew:

[
H_{F} : Inv(M^{k}) ⊆ Inv(M^{a}) ;\wedge; Inv(M^{k}) ⊆ Inv(M^{\ell}) ;\wedge; Inv(M^{a}) ⊆

$$
Inv(M^{\ell})
$$

- **Two-Source flavour (Mark + Q)**
  There exists a latent set Q such that:

$$
Inv(M^{a}) = Inv(M^{k}) \cup Q \cup M \quad\text{and}\quad Inv(M^{\ell}) = Inv(M^{k}) \cup Q \cup L
$$

where M and L are special material in Matthew and Luke.

In practice, we cannot yet directly compute ⊆ across entire sets; but we can look for patterns where one Gospel's pericopes form a **minimal spanning set** of invariants that others elaborate.

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5. Results: The First λ-Coherence Snapshot

Running the comparison script over the first sample yields the following pattern of similarities (rounded to three decimal places):

- **Calming the Storm** (Matt 8:23–27; Mark 4:35–41; Luke 8:22–25)
  - sim(Matt, Mark) ≈ **0.523**
  - sim(Matt, Luke) ≈ **0.538**
  - sim(Mark, Luke) ≈ **0.841**
- **Gerasene demoniac** (Matt 8:28–34; Mark 5:1–20; Luke 8:26–39)
  - sim(Matt, Mark) ≈ **0.468**
  - sim(Matt, Luke) ≈ **0.493**
  - sim(Mark, Luke) ≈ **0.900**
- **Jairus & bleeding woman** (Matt 9:18–38; Mark 5:21–43; Luke 8:40–56)
  - sim(Matt, Mark) ≈ **0.320**
  - sim(Matt, Luke) ≈ **0.329**
  - sim(Mark, Luke) ≈ **0.965**
- **Feeding of the 5,000** (Matt 14:13–21; Mark 6:30–44; Luke 9:1–17)
  - sim(Matt, Mark) ≈ **0.514**
  - sim(Matt, Luke) ≈ **0.486**
  - sim(Mark, Luke) ≈ **0.741**
- **Peter's confession** (Matt 16:13–20; Mark 8:27–38; Luke 9:18–27)
  - sim(Matt, Mark) ≈ **0.392**

    - sim(Matt, Luke) ≈ **0.379**
    - sim(Mark, Luke) ≈ **0.960**
- **Transfiguration** (Matt 17:1–13; Mark 9:2–13; Luke 9:28–36)
    - sim(Matt, Mark) ≈ **0.434**
    - sim(Matt, Luke) ≈ **0.442**
    - sim(Mark, Luke) ≈ **0.703**
- **Empty tomb** (Matt 28:1–10; Mark 16:1–8; Luke 24:1–12)
    - sim(Matt, Mark) ≈ **0.372**
    - sim(Matt, Luke) ≈ **0.461**
    - sim(Mark, Luke) ≈ **0.445**

Even in this small sample, a *consistent pattern* emerges:

1. **Mark–Luke λ-similarity is often highest** in strongly narrative episodes, especially:
    - Gerasene demoniac
    - Jairus & bleeding woman
    - Feeding of the 5,000
    - Peter's confession
2. Matthew's λ-invariants, as currently encoded, tend to be slightly further from Mark and Luke. This matches Matthew's reputation as the more **didactically structured and ecclesially inflected** Gospel, where the same event is often embedded in a broader teaching block.
3. In the **rich young ruler** and **servant-ransom saying**, the current encoding yields low similarities for Matthew compared to Mark and Luke. This reflects both:
    - genuine theological emphases (Matthew's focus on reward, reversal, and the first/last motif), and
    - our conscious choice to encode Matthew's invariants slightly more "programmatically" than Mark's.

In other words: even with a simple string-based metric, **Mark and Luke cluster tightly** in many of the high-stakes narrative pericopes, with Matthew often acting as the "editorial theologian" whose λ-formulas sit one step further out.

This is **not a proof** of Markan priority. But it does show that when we ask, "Which Gospel's λ-invariants form the most reusable skeleton for the others?" the answer in these cases is **closer to Mark's narrative λ-forms than to Matthew's or Luke's fuller discourses.**

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6. Interpreting λ-Coherence

6.1 What λ-Coherence Is *Not*

First, guardrails:

- It is **not** a claim that we have discovered the "true source" behind the Synoptics.
- It is **not** a statistical argument in the sense of large-scale corpus linguistics; the sample is small and the encoding is hand-crafted.
- It is **not** a replacement for redaction-critical or tradition-historical work, but a scaffold on which those arguments can now be **formalised and tested**.

6.2 What λ-Coherence *Does* Offer

What λ-coherence *does* give us is:

1. A **precise, machine-checkable representation** of what each interpreter (in this case, the iota Verbum engine plus its human partners) takes to be the **canonical invariant** of a pericope.
2. A way to spot **which Gospels, under a shared grammar, tend to agree at the level of invariant structure** rather than just phrasing.
3. A way to phrase hypotheses like "Mark is the minimal λ-skeleton" or "Luke expands a Mark + Matthew mix" as concrete constraints on sets of λ-formulas, rather than as vague statements about "dependence" or "influence."

In this first snapshot, the strong Mark–Luke pairing in several pericopes is exactly what you would expect if:

- Mark's λ-forms capture a **short, vivid skeleton** of events, and
- Luke's λ-forms aim to be historically anchored narratives that track that skeleton quite closely, even while re-contextualising them for Theophilus and Gentile audiences.

Matthew, by contrast, often **bundles events into larger teaching units**, so his λ-formulas in our encoding sometimes fold multiple narrative effects into one invariant.

6.3 Modal Hypotheses in Practice

Given the data, we can already start to phrase tests like:

- **Hypothesis:** For triple tradition pericopes, *if* Mark's λ-form is treated as the base, *then* both Matthew's and Luke's λ-forms can be seen as **□-preserving expansions**:

$$
[
\forall p \in \text{TripleTradition}:
\quad λ_{Mark}(p) ⊢ λ_{Matthew}(p) ;\wedge; λ_{Mark}(p) ⊢ λ_{Luke}(p)
]
$$

where ⊢ is logical consequences in the modal calculus.

- **Checkpoint:** Our similarity scores suggest that for many such p, sim(Mark, Luke) is sufficiently high that this kind of entailment is at least **plausible** with moderate editing.

Similarly, we can encode a **Farrer-style** test:

- For double tradition material, if Luke is using both Mark and Matthew, Luke's λ-form should often be a **least common super-invariant** of the others.

These tests are not yet implemented in full theorem-proving style, but the λ-documents and comparison scripts we already have make this future work extremely concrete.

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7. Canonical & Theological Reflection

7.1 One Gospel, Three λ-Projections

From a canonical standpoint, the point of λ-coherence is not to collapse the three Gospels into one bland "harmonised" account. It is to make visible:

- The **shared invariant core** of the Gospel: who Jesus is, what he does, what his death and resurrection accomplish.
- The **distinct ways** each Evangelist projects that core into their community's needs.

In this frame:

- **Mark** often appears as the *crisis-mode λ-projection*: fast, urgent, vivid; a skeleton that can be retold under persecution.
- **Matthew** appears as the *catechetical λ-projection*: structured discourse, Torah fulfilment, church instruction.
- **Luke** appears as the *historical-pastoral λ-projection*: careful compilation, attention to the marginalised, bridge to Acts.

λ-coherence does not abolish these differences; it confirms that beneath them lies a **shared constraint set**—a canonical charter—about Jesus' identity and mission.

7.2 Trust, Attestation, and the Gospels as Charter

There is a deeper connection to iota Verbum's broader mission.

Later, the same modal engine is applied to **institutions**:

- A bank expresses its **charter** (who it claims to be) in modal form.
- Each credit decision is checked and **attested** against that charter.
- Over time, the institution's actual behaviour is compared with its professed identity.

In a sense, the Synoptic Gospels are already this: four attestations of **one divine identity**, rendered under pressure, with different emphasised contexts.

By making the Synoptics executable in λ-form, we are:

- Treating them as a **reference implementation** of a trust system:
  the strongest possible testbed for a logic that must handle ambiguity, suffering, injustice, and apparent contradiction without losing identity coherence.
- Learning the **shape of faithfulness under pressure** in logic form, before asking institutions or AI systems to do anything analogous.

8. Implications and Future Work

8.1 For New Testament Studies

For biblical scholarship, the value of this approach is **not** that it "solves" the Synoptic Problem once and for all, but that it:

1. Makes our **interpretive choices explicit** in code (what we think the invariant of each pericope is).
2. Enables repeatable **experiments** on how hypotheses about dependence fare under those choices.
3. Offers a platform where different scholars could **encode rival λ-corpora** and see how their Synoptic hypotheses pan out.

Future work could include:

- Extending λ-encoding to **all pericopes** in Matthew, Mark, and Luke (we already have full coverage for Matthew, a strong sample for Mark, and a key set for Luke).
- Introducing **semantic embeddings** or type systems, so similarity is measured not just by string overlap but by deeper relational structure.
- Using theorem-proving or SAT-style tools to check whether particular hypothesis families (Two-Source, Farrer, etc.) can be made globally consistent with a given λ-corpus.

8.2 For Computational Theology

For computational theology, the Synoptic λ-corpus is:

- A **living dataset** that encodes a canonical text as a graph of modal invariants.
- A template for encoding other corpora (e.g. Pauline letters, Psalms) as compatible λ-slices.
- A bridge between **formal methods** and **dogmatic or biblical theology**: the same tools that verify hardware or protocols are repurposed to ask whether our reading of Scripture is internally coherent.

8.3 For Trust Infrastructure and AI

Finally, for AI governance:

- The Synoptic λ-coherence experiment demonstrates that **one modal grammar** can both:
    - handle a complex, debated ancient text under interpretive pressure, and
    - later handle **high-stakes institutional decisions**.
- This cross-domain portability is precisely what you want in a **trust substrate**: not just another metrics dashboard, but a logic whose DNA has been trained on long-argued questions of identity, justice, mercy, and coherence under suffering.

The point is not that a bank is "like" a Gospel, but that both are subject to a question λ-logic can formalise:

Are you acting in a way that is recognisably faithful to who you say you are?

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9. Conclusion

The Synoptic Problem will not be "solved" by one more paper, and certainly not by a single coding experiment. But **Synoptic λ-coherence** does something new:

- It shows that we can encode Gospel pericopes as **executable invariants** in a shared modal grammar.
- It shows that we can **measure and visualise** how closely those invariants align across Matthew, Mark, and Luke.
- It suggests that **Mark's narrative λ-forms often function as a minimal skeleton** that Luke closely tracks and Matthew re-projects in didactic form—a result broadly compatible with Markan-priority families of solutions, though not dictated by them.
- It provides a **working test harness** in which future hypotheses—Two-Source, Farrer, Augustinian, or completely new models—can be expressed as constraints on λ-corpora and evaluated in a transparent way.

Most importantly for iota Verbum, it proves that the **same modal engine** that can faithfully respect the diversity and unity of the Gospel story can also serve as an **outer conscience layer** for AI and institutions. If a logic can survive the Synoptics, it is at least a candidate for the kind of trust we now need in other domains.

The next steps are clear:

1. Complete λ-encoding for all Synoptic pericopes.
2. Tighten the similarity and entailment machinery.
3. Invite other scholars and practitioners to contribute their own λ-encodings and hypotheses.

The Synoptic λ-corpus thus becomes both a **gift back to theology** and a **foundation for trustworthy computation**. One Gospel in three projections, one modal grammar, many domains.